A Note on the Frohmanian Symplectic Tether Framework Applied to a Functional Equation

Benjamin Frohman

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Abstract

We illustrate how the methodological tools of the Frohmanian Symplectic Tether Theorem — uniqueness via canonicity conditions, invariance under iteration (coadjoint analogue), reduced-orbit necessity, and independent majorant control — directly yield clean first-principles proofs of the three statements in problem B4 concerning continuous real-valued functions f satisfying $f(x^2) = f(x)^2$.

1 Setup

Let $f : D \to \mathbb{R}$ be continuous and satisfy the multiplicative invariance

$$f(x^2) = f(x)^2 \quad \forall x \in D.$$

Define the (forward/backward) orbit of a point x under the iteration map $\phi(y) = y^2$. By continuity, behavior along orbits is controlled. This invariance is analogous to coadjoint invariance in the $\mathfrak{sdiff}(\mathbb{T}^3)$ setting.

2 Proof of (i): Uniqueness on $[0, 1]$ with $f(0) \neq 0$

Consider any $x \in [0, 1]$. The forward orbit satisfies $x^{2^n} \to 0$. Then

$$f(x^{2^n}) = f(x)^{2^n}.$$

Taking $n \to \infty$, continuity implies

$$f(0) = \lim_{n\to\infty} f(x)^{2^n}.$$

Assume $f(0) \neq 0$. Suppose there exists x with $f(x) \neq 1$: - If $|f(x)| > 1$, then $|f(x)^{2^n}| \to \infty$, contradicting continuity at 0. - If $|f(x)| < 1$, then $f(x)^{2^n} \to 0$, again contradicting $f(0) \neq 0$.

Thus $f(x) = 1$ (sign consistency follows from squaring). By density of orbits in $[0, 1]$ under iteration and continuity, $f \equiv 1$ on $[0, 1]$.

This mirrors the **5-step canonicity** (C1–C5) and uniqueness theorem: the invariance forces a unique canonical solution, with degeneracy on constant proxies ruling out alternatives.

3 Proof of (ii): All-or-Nothing Non-Vanishing on $(0, \infty)$

Suppose there exists $a > 0$ with $f(a) = 0$. Then $f(a^{1/2^n}) = 0$ for all n. The backward orbit $a^{1/2^n} \to 1$, so continuity yields $f(1) = 0$.

From $f(1) = 0$, forward iteration and density of orbits under roots/powers fill $(0, \infty)$ by continuity. Thus $f \equiv 0$.

Conversely, non-vanishing at one point propagates everywhere.

This is the direct analogue of **reduced-orbit necessity** (T3 shear counterexample): without the proper reduction (connected components under the iteration map), global invariance fails.

4 Proof of (iii): Infinitely Many f on $(0, \infty)$ with $\int_0^\infty f(x)\,dx < 1$

On $(1, \infty)$, prescribe f freely on a fundamental domain (intervals not in the image of squaring from below) and extend via $f(x) = \pm\sqrt{f(x^2)}$ (continuous branch). Choose decay near 0 and ∞ (e.g., $f(x) \sim x^r$ for $r > 0$, or localized bumps) so the integral is finite and arbitrarily small.

The freedom on the fundamental domain yields infinitely many distinct solutions.

This follows the **independent majorant + negative quadratic feedback** philosophy: the squaring relation plus continuity provides controllable decay (quartic absorption analogue), while the tether proxy in the discrete setting mirrors the freedom under global constraints.

5 Conclusion: Direct Methodological Transfer

The Frohmanian framework supplies the precise toolkit:

- Uniqueness/canonicity under forcing conditions (parts (i)–(ii)).
- Orbit reduction and invariance (coadjoint analogue).
- Independent majorant control for global bounds (part (iii)).

No ansatz is required; all results follow from minimal geometric/functional constraints + continuity, exactly as in the tether derivation for global Navier–Stokes regularity.